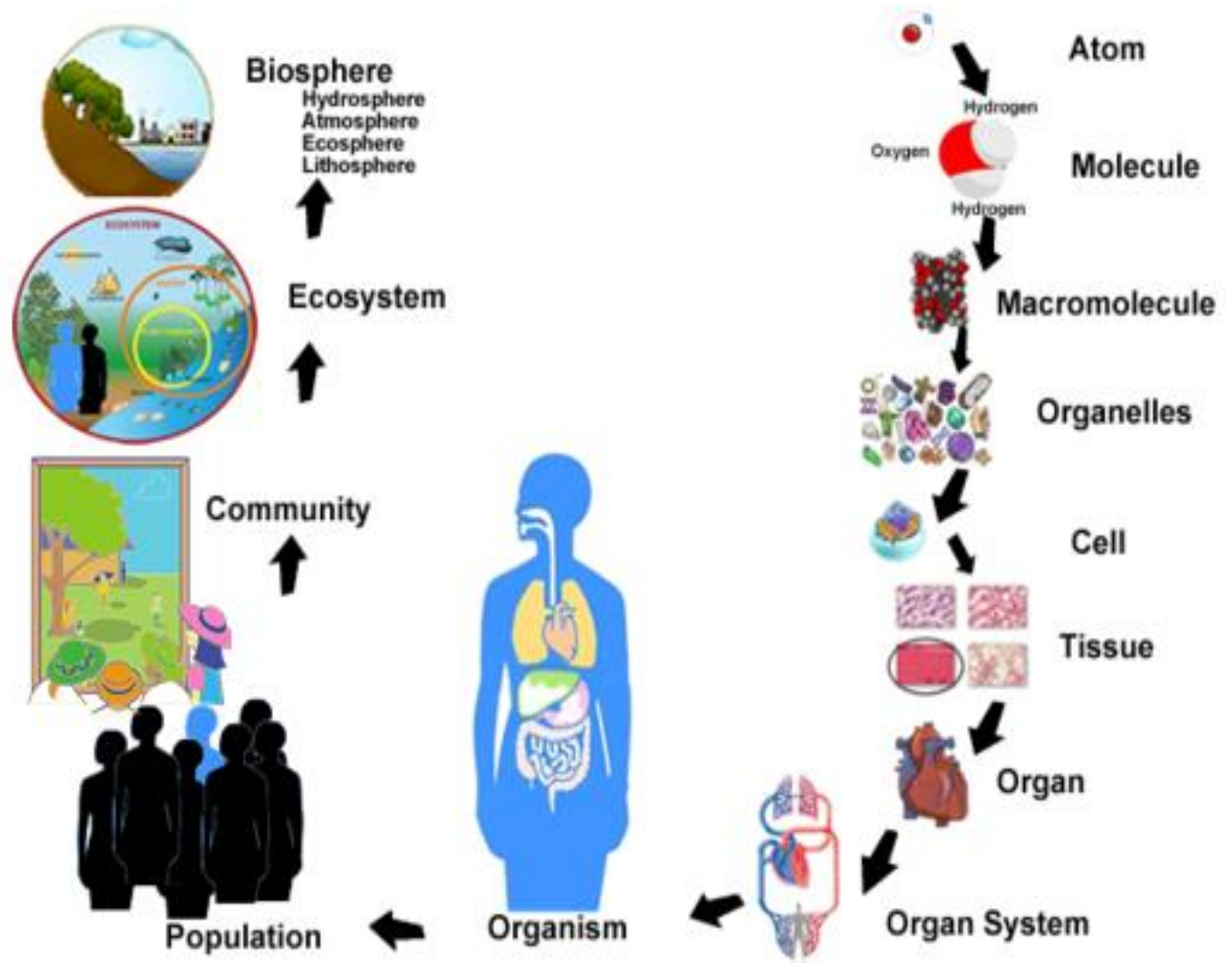


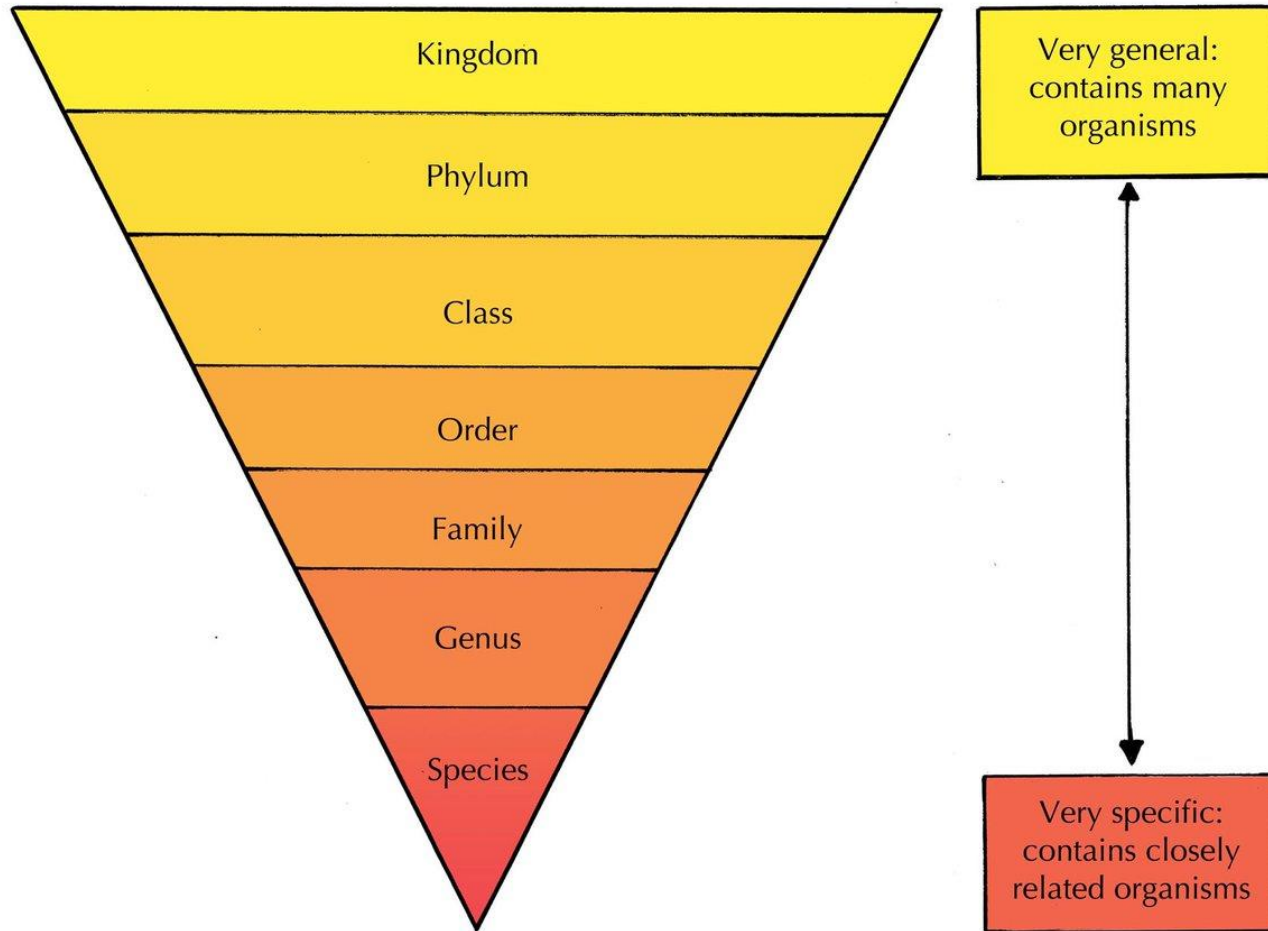
The biological levels of organization

The biological levels of organization of living things arranged from the simplest to most complex are: atom, molecules, macromolecules, organelles, cells, tissues, organs, organ systems, organisms, populations, communities, ecosystem, and biosphere.

Living things are highly organized and structured, following a hierarchy that can be examined on a scale from small to large.



Taxonomic_Hierarchy



kingdom - **animalia**
phylum - **chordata**
class - **mammalia**
order - **carnivorora**
family - **felidae**
genus – **panthera**
species - **tigris**

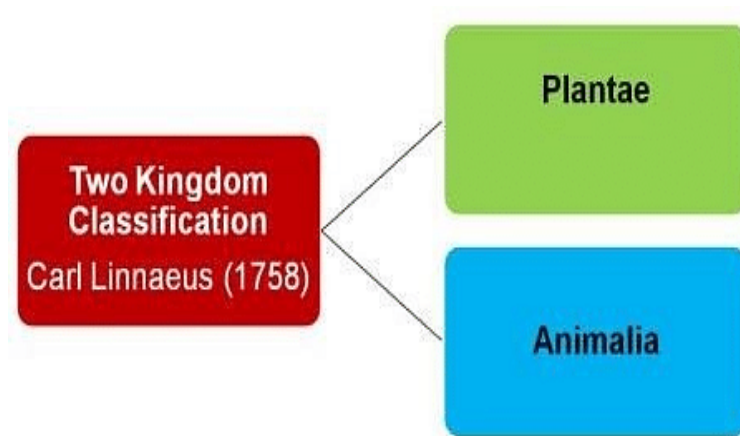


Biological classification

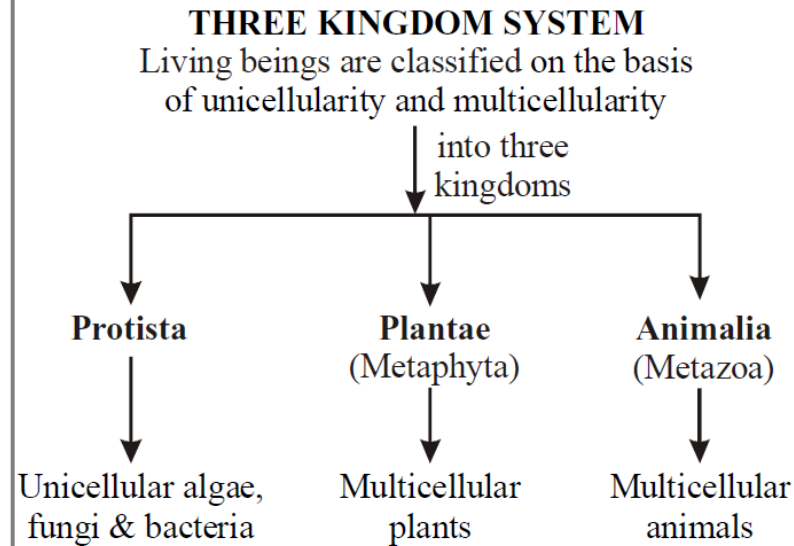
The classification systems have undergone several changes with time. The characteristics of organisms vary greatly. The structure, form, life cycle, and metabolism of all living organisms differ. These numerous organisms are divided into groups for easy identification and comprehension.

The first attempt of classification was made by **Aristotle (Father of Biology)**. He classified living organisms based on their ability to move from one place to another or **locomote**. This gave rise to **two groups, plants which are non-motile and animals which are motile**. He used simple morphological features to further classify plants into herbs, shrubs and trees and classified animals on the basis of presence and absence of RBC. He also made an attempt in classifying different organisms based on where they live - land, water, or air. This system of classification failed to classify all the known organisms.

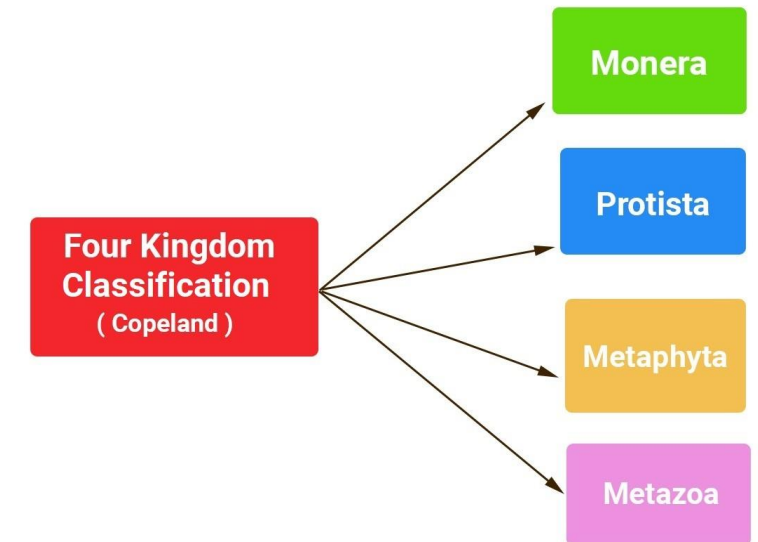
Two Kingdom classification



Three Kingdom classification

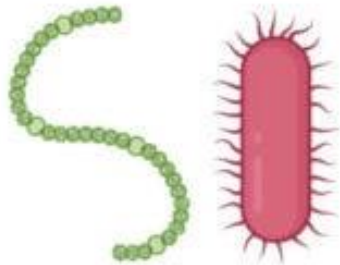


Four Kingdom classification

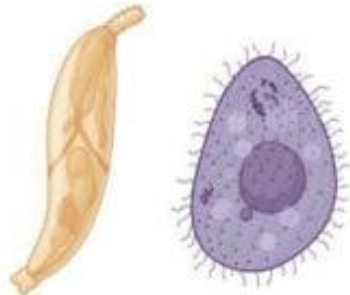


Five Kingdom Classification

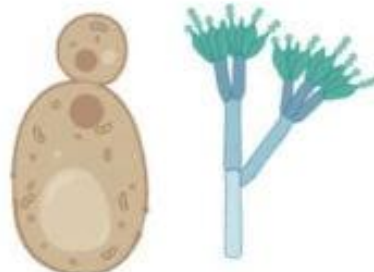
Monera



Protist



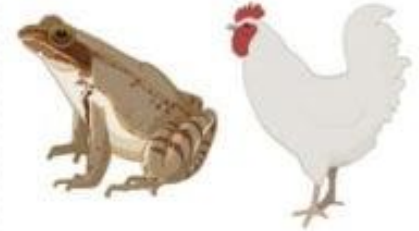
Fungi



Plantae

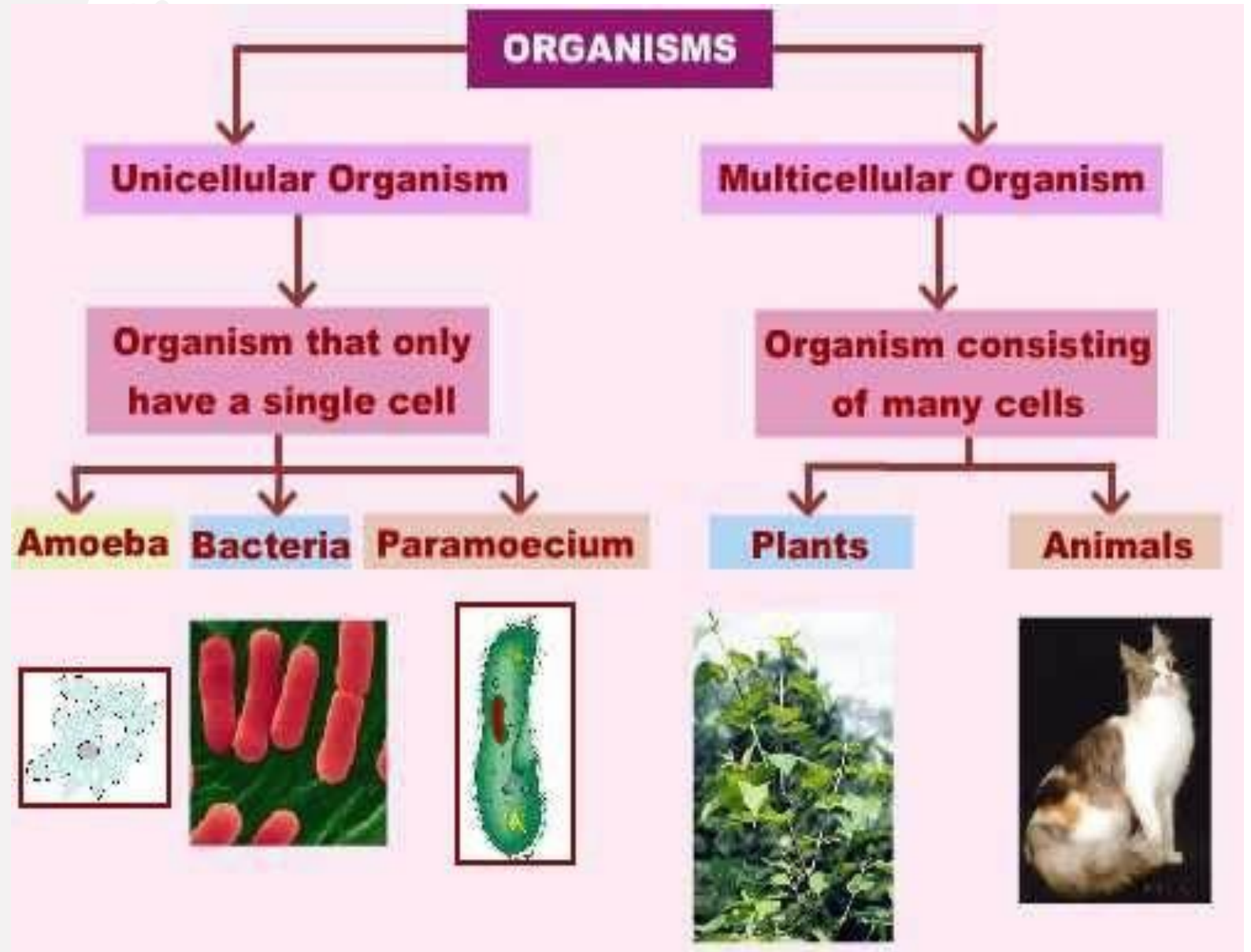


Animalia



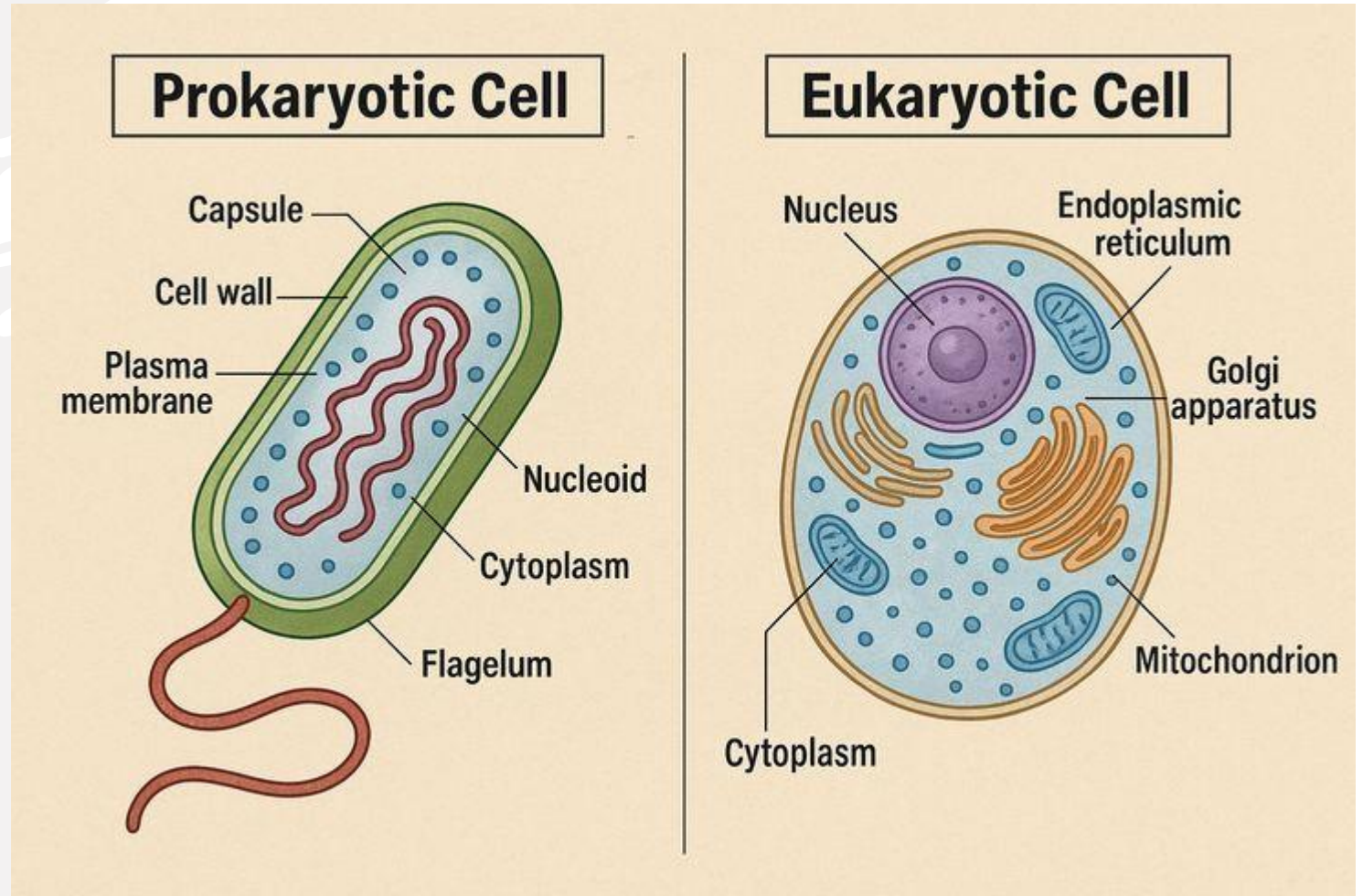
Classification Based On Cellularity

Based on the number of cells the organisms possess, they can be classified into **Unicellular Organisms** and **Multicellular Organisms**.



Classification Based On Ultrastructure

Based on the Ultrastructure of cell, the organisms can be classified into **prokaryotes** (lacking a membrane-bound nucleus and organelles) and **eukaryotes** (possessing a complex, compartmentalized cell structure)



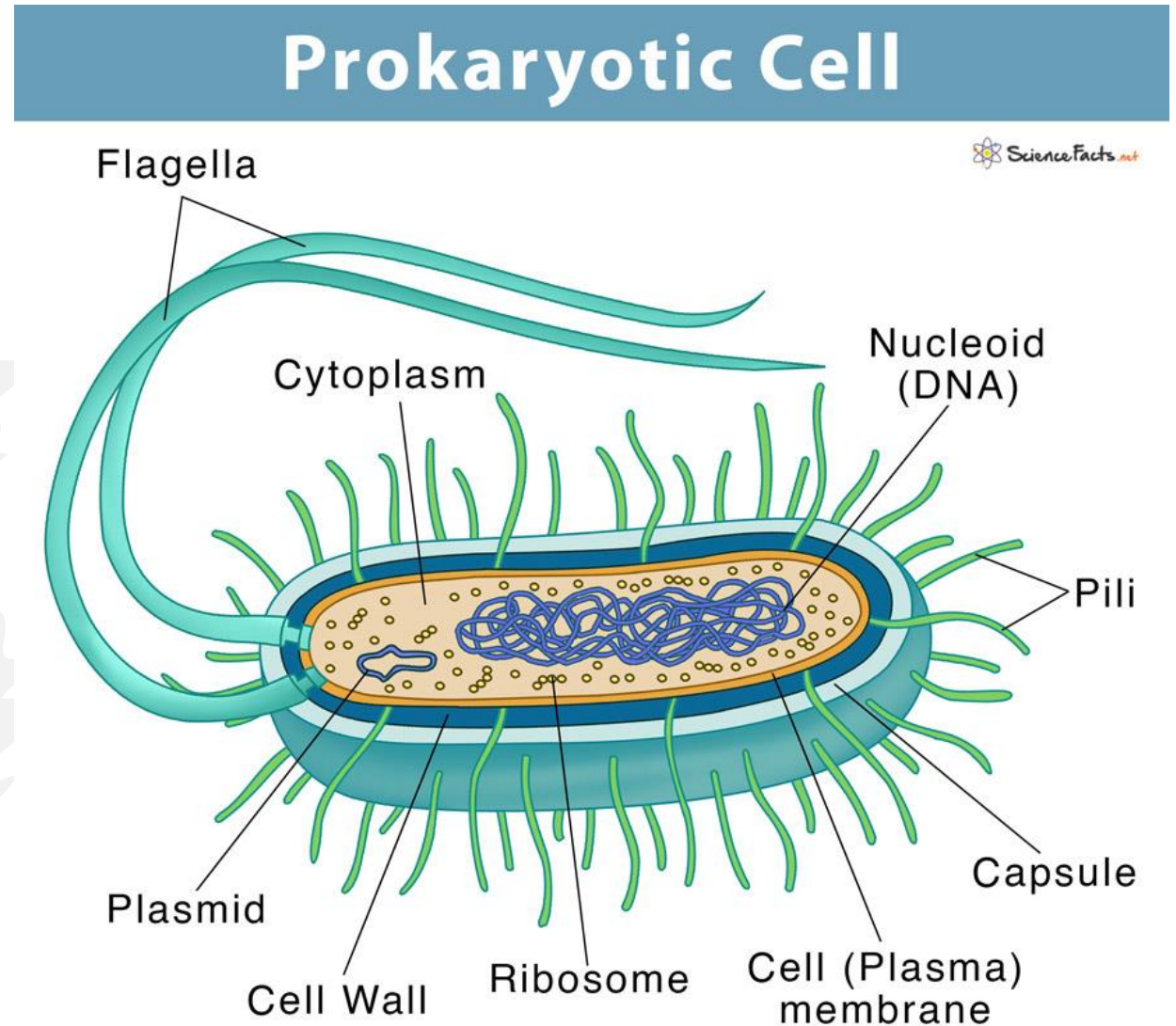
Based on the ultrastructure of cell, the organisms can be classified into Prokaryotic Organisms and Eukaryotic Organisms.

Prokaryotic cells are single-celled microorganisms known to be the earliest on earth. Prokaryotes include Bacteria and Archaea. The photosynthetic prokaryotes include cyanobacteria that perform photosynthesis.

Prokaryotic Cell Structure

A prokaryotic cell does not have a nuclear membrane. However, the [genetic material](#) is present in a region in the cytoplasm known as the nucleoid. A prokaryotic cell lacks membrane bound cell-organelles like mitochondria, endoplasmic reticulum, and Golgi bodies.

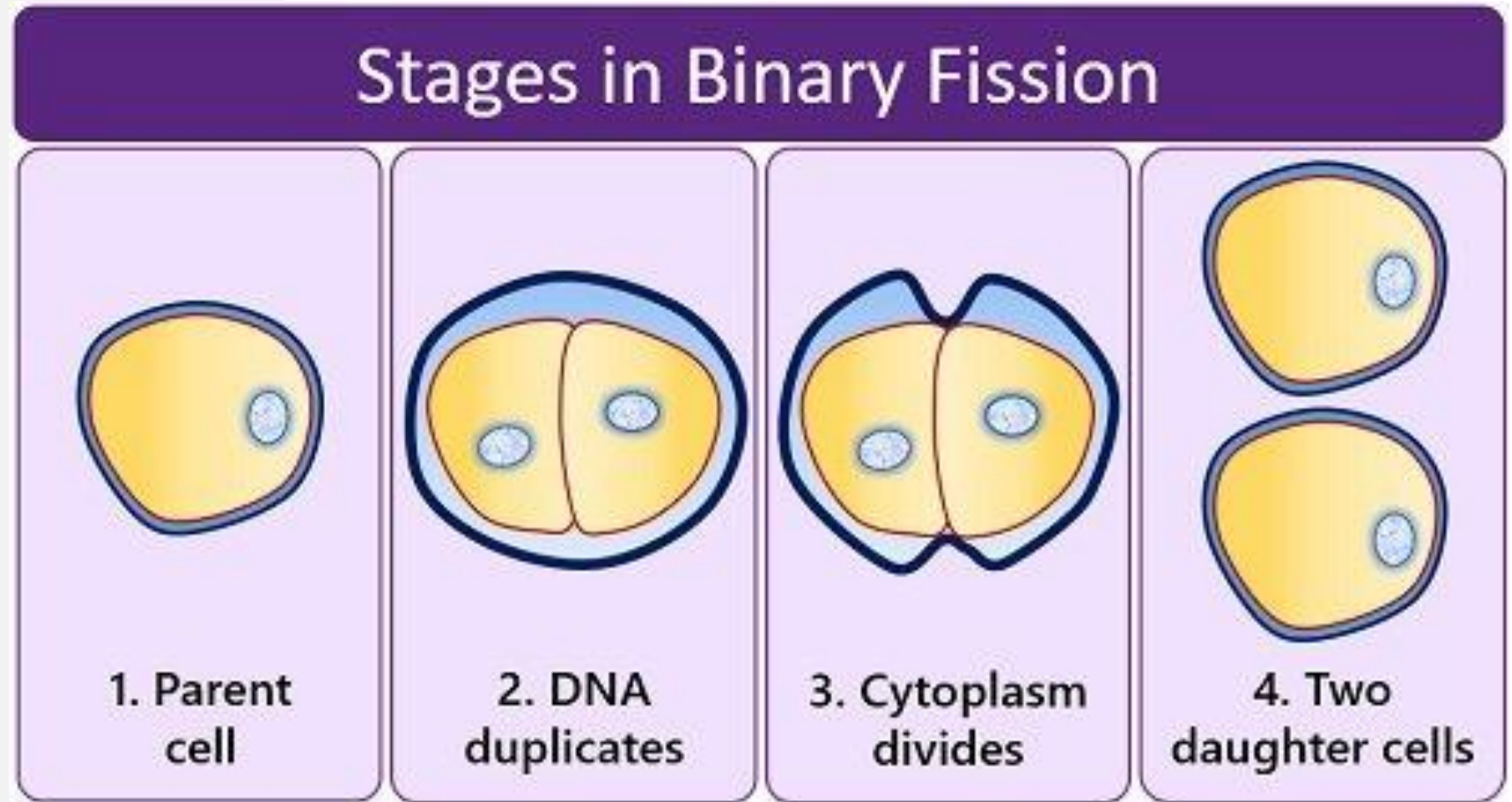
They may be spherical, rod-shaped, or spiral. A prokaryotic cell structure is as follows:



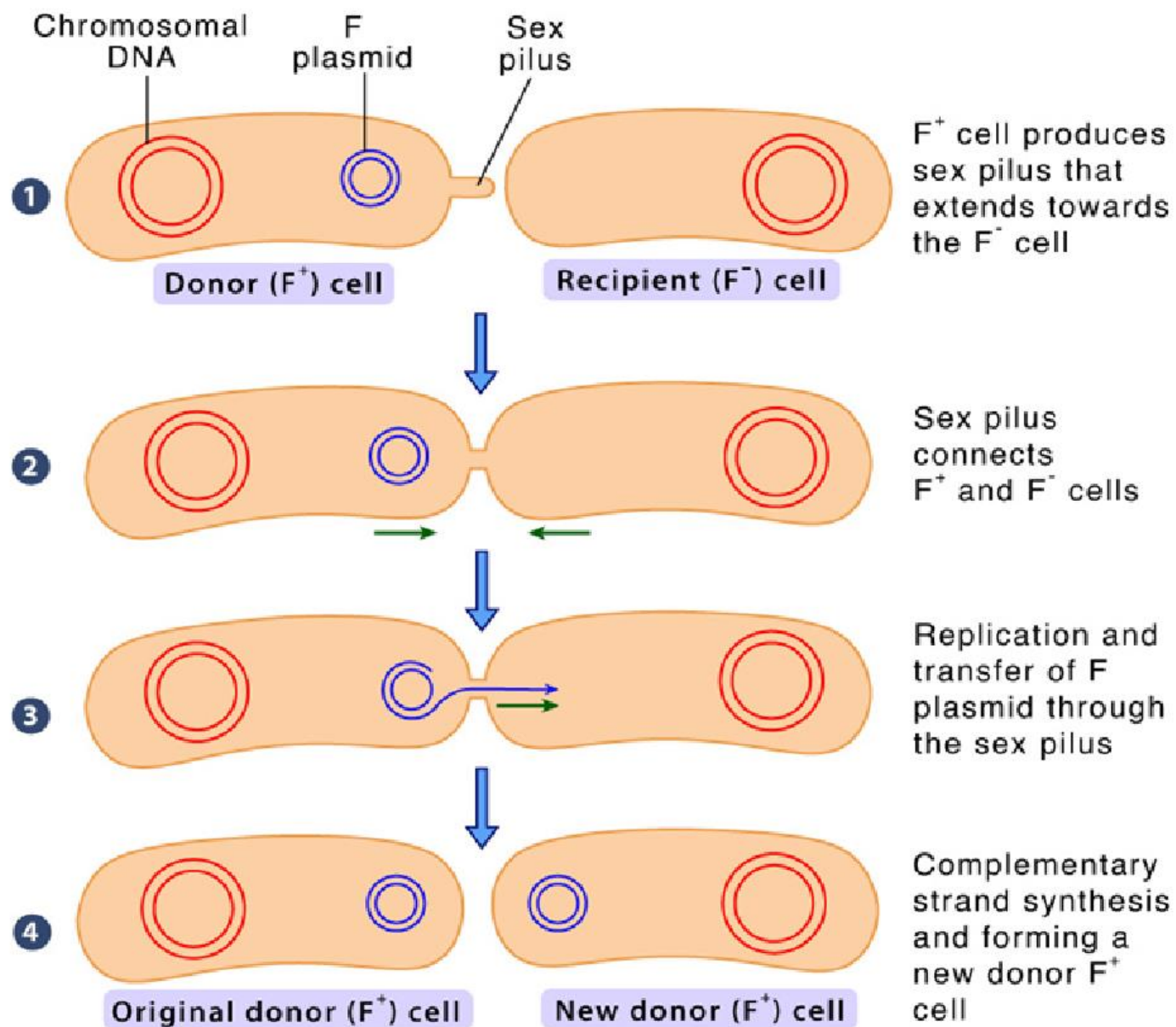
- **Capsule**– It is an outer protective covering found in the bacterial cells, in addition to the cell wall. It helps in moisture retention, protects the cell when engulfed, and helps in the attachment of cells to nutrients and surfaces.
- **Cell Wall**– It is the outermost layer of the cell which gives shape to the cell.
- **Cytoplasm**– The cytoplasm is mainly composed of enzymes, salts, cell organelles and is a gel-like component.
- **Cell Membrane**– This layer surrounds the cytoplasm and regulates the entry and exit of substances in the cells.
- **Pili**– These are hair-like outgrowths that attach to the surface of other bacterial cells.
- **Flagella**– These are long structures in the form of a whip, that help in the locomotion of a cell.
- **Ribosomes**– These are involved in protein synthesis.
- **Plasmids**– Plasmids are non-chromosomal DNA structures. These are not involved in reproduction.
- **Nucleoid Region**– It is the region in the cytoplasm where the genetic material is present.

Reproduction in Prokaryotes

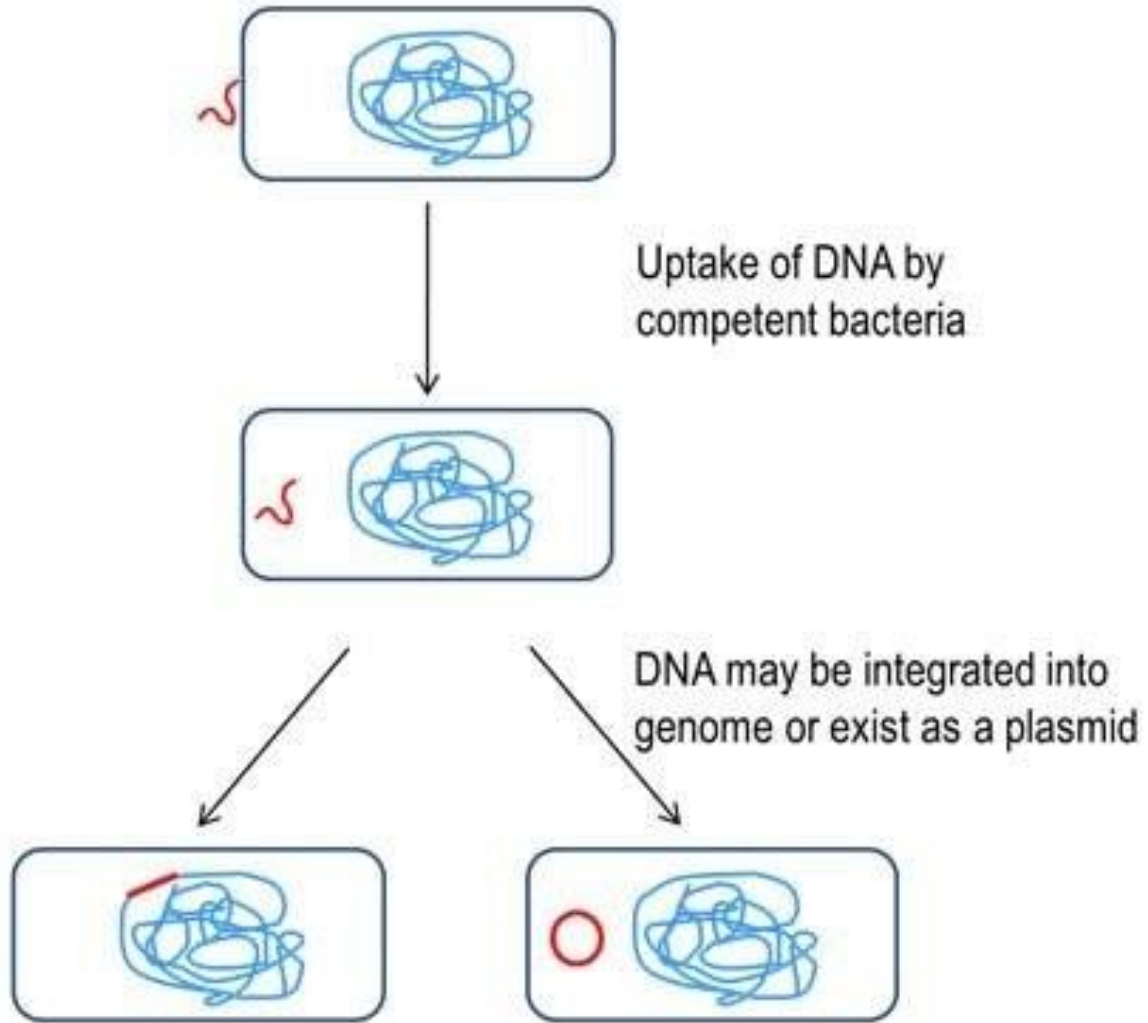
Prokaryotes reproduce asexually by **binary fission**; they can also exchange genetic material by **transformation**, **transduction**, and **conjugation**.



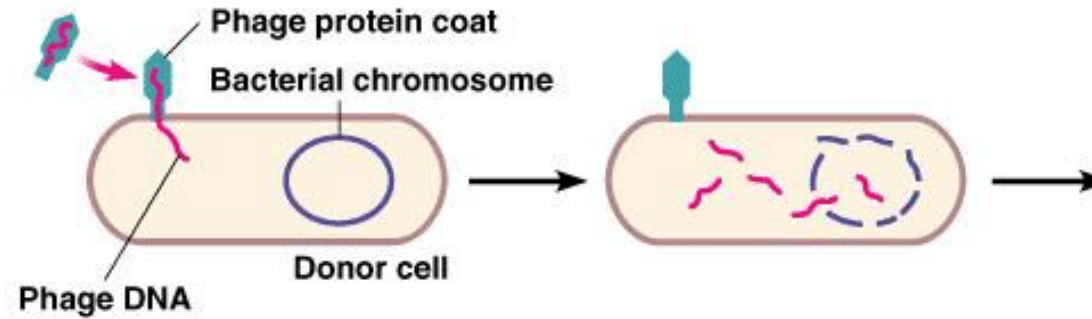
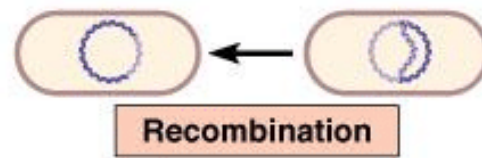
Conjugation



Transformation

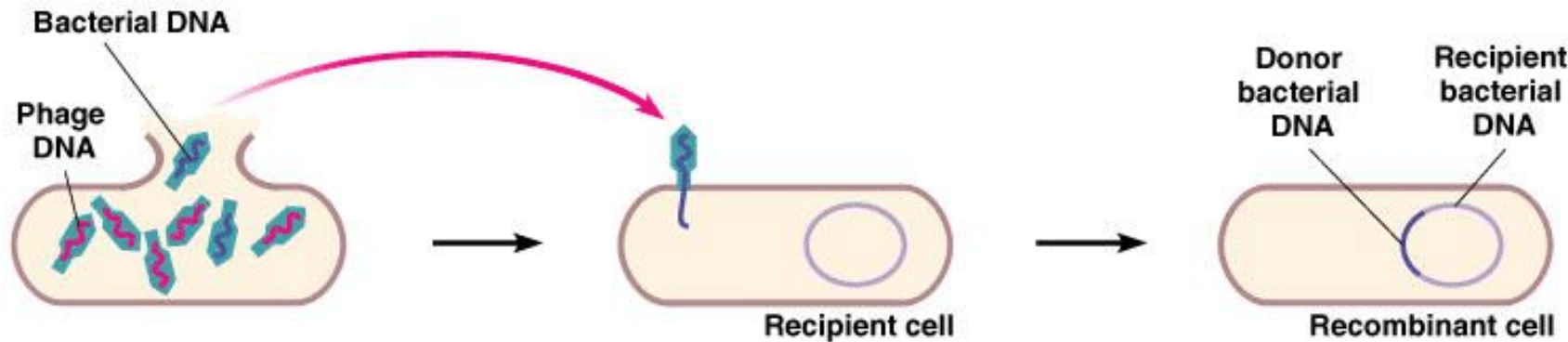


Transduction



1 A phage infects the donor bacterial cell.

2 Phage DNA and proteins are made, and the bacterial chromosome is broken down into pieces.



3 Occasionally during phage assembly, pieces of bacterial DNA are packaged in a phage capsid. Then the donor cell lyses and releases phage particles containing bacterial DNA.

4 A phage carrying bacterial DNA infects a new host cell, the recipient cell.

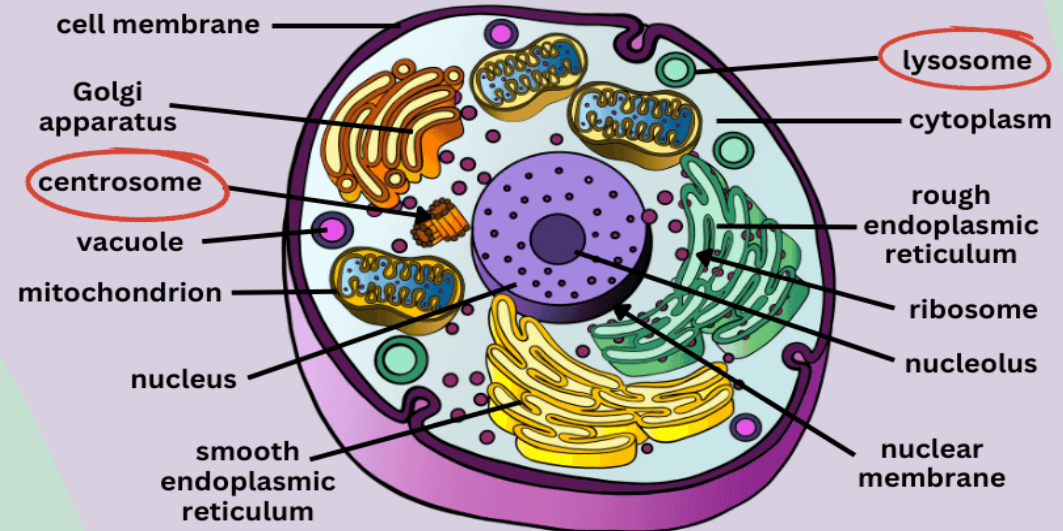
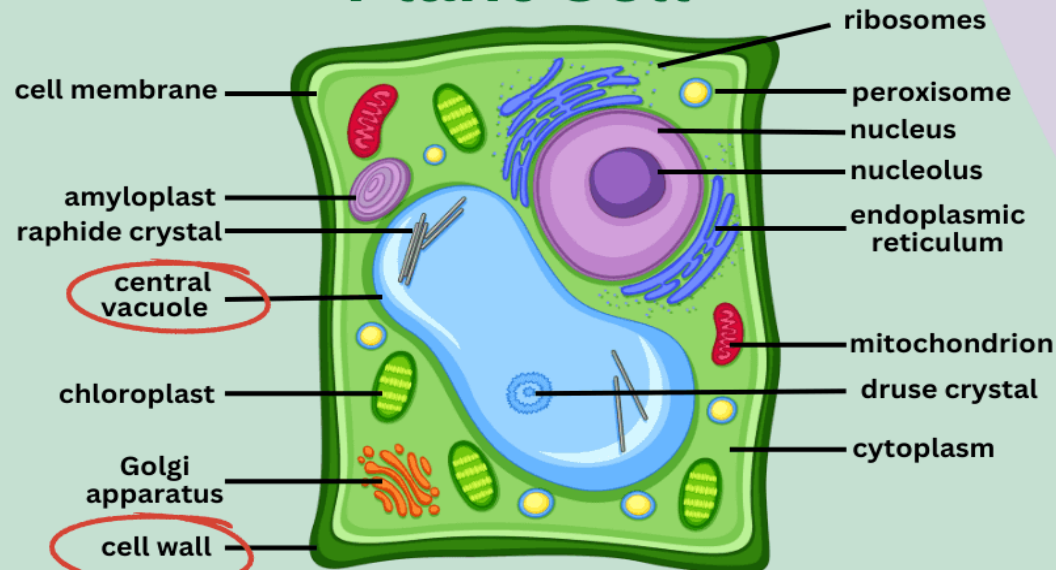
5 Recombination can occur, producing a recombinant cell with a genotype different from both the donor and recipient cells.

Eukaryotic Cell

PLANT CELL VS ANIMAL CELL

- Cell wall (rigid shape)
- Large central vacuole
- Chloroplasts (autotroph)
- Lack centrosomes
- Lack lysosomes

Plant Cell



Animal Cell

- No cell wall (irregular shape)
- Numerous small vacuoles
- No chloroplasts (heterotroph)
- Centrioles / centrosomes
- Lysosomes

Structure Of Eukaryotic Cell

The eukaryotic cell structure comprises the following:

Plasma Membrane

- The plasma membrane separates the cell from the outside environment.
- It comprises specific embedded proteins, which help in the exchange of substances in and out of the cell.

Cell Wall

- A cell wall is a rigid structure present outside the plant cell. It is, however, absent in animal cells.
- It provides shape to the cell and helps in cell-to-cell interaction.
- It is a protective layer that protects the cell from any injury or pathogen attacks.
- It is composed of cellulose, hemicellulose, pectins, proteins, etc.

Cytoplasm

The cytoplasm is the gel-like fluid inside the cell. It is the medium for chemical reaction. It provides a platform upon which other organelles can operate within the cell. All of the functions of cell are carried out in the cytoplasm.

Endoplasmic Reticulum

It is a network of small, tubular structures that divides the cell surface into two parts: luminal and extraluminal.

Endoplasmic Reticulum are of two types:

- Rough Endoplasmic Reticulum contains ribosomes.**
- Smooth Endoplasmic Reticulum that lacks ribosomes and is therefore smooth.**

Nucleus

- The nucleoplasm enclosed within the nucleus contains DNA and proteins.**
- The nuclear envelop consists of two layers- the outer membrane and the inner membrane. Both the membranes are permeable to ions, molecules, and RNA material.**
- Ribosome production also takes place inside the nucleus.**

Golgi Apparatus

- It is made up of flat disc-shaped structures called cisternae.**
- It is absent in red blood cells of humans and sieve cells of plants.**
- They are arranged parallel and concentrically near the nucleus.**
- It is an important site for the formation of glycoproteins and glycolipids.**

Ribosomes

These are the main site for protein synthesis and are composed of proteins and ribonucleic acids.

Mitochondria

- **These are also known as “powerhouse of cells” because they produce energy.**
- **It consists of an outer membrane and an inner membrane. The inner membrane is divided into folds called cristae.**
- **They help in the regulation of cell metabolism.**

Lysosomes

They are known as “suicidal bags” because they possess hydrolytic enzymes to digest protein, lipids, carbohydrates, and nucleic acids.

Plastids

These are double-membraned structures and are found only in plant cell. These are of three types: Chloroplast, Chromoplast and Leucoplasts.

Vacuole

A vacuole is a membrane bound structure found in the cytoplasmic matrix of a cell. They can store food or other nutrients required by a cell to survive. They contain anthocyanins which are water-soluble and are responsible for pink, red, purple, and blue in flowers, vegetables, and fruits. The vacuoles in plant cells are larger than those in the animal cells.

Classification Based On Energy & Carbon use

Food is the only source of energy for all living organisms on this planet. This food is available from different sources. Organisms are characterized into two broad categories based upon how they obtain their energy and nutrients:

- **Autotrophs**
- **Heterotrophs**
- **Lithotrophs**

Autotrophs: Autotrophs are organisms who utilise sun's energy and simple compounds like carbon dioxide and water to store energy in food (glucose and starch). They are not dependent on other organisms for energy and instead directly obtain it from sunlight. Because of their ability to make their own food, autotrophs are also commonly referred to as **primary producers** and thus occupy the base of the food chain. They vary widely from those found on land (soil) to those that live in aquatic environments.

Example: Most of the plants, Cyanobacteria, Seaweed, Phytoplankton etc.



Heterotrophs: Heterotrophs are those organisms that cannot prepare their own food and depend on other organisms (plants or animals) for food. They are directly or indirectly dependent on these producers for food. Heterotrophs occupy the second and third levels in a food chain, a sequence of organisms that provide energy and nutrients for other organisms.

Example: All animals, human beings, fungi etc.

Lithotrophs: Lithotrophs are microorganisms that use **inorganic substrates** as a source of electron donors to drive energy acquisition, using either organic carbon or carbon dioxide as a source of carbon for constructing cellular materials. Lithotrophs are found in soil and aquatic environments wherever their energy source is present. Most lithotrophs are autotrophs so they can grow in the absence of any organic material.

Example: nitrifying bacteria that use ammonia and convert it into nitrite or use nitrite to produce nitrate, hydrogen bacteria that oxidize hydrogen to water etc



Classification Based On excretion

Metabolism produces different types of waste products. Digestion and catabolism of proteins and nucleic acids mainly result in nitrogenous waste. **According to the type of nitrogenous waste animals excrete, there are three groups of animals as Ammonotelic, ureotelic, uricotelic and Aminotelic.**

Ammonotelic organism:

- An ammonotelic organism generally **excretes nitrogenous waste as soluble ammonia**.
- Ammonia, the product or waste formed is **highly toxic** and requires a **large amount of water for its excretion**.
- Most aquatic animals including protozoans, crustaceans, Platyhelminthes, cnidarians, poriferans, echinoderms, fishes, larvae/tadpoles of amphibians are ammonotelic.

Ureotelic organism:

- A ureotelic organism generally excretes excess **nitrogen as urea**. Urea is **less toxic** and needs **less water for excretion**.
- Ureotelic organisms include cartilaginous fish, a few bony fishes, adult amphibians, and also mammals including humans.

Uricotelic organism:

- A uricotelic organism generally excretes excess **nitrogen as uric acid**.
- The excreta of the uricotelic organisms are **insoluble solids or semi-solid** and **requires less water**.
- For example: Terrestrial arthropods (including insects), lizards, snakes, birds, etc.

Aminotelic organism:

- The organisms which excrete nitrogenous waste **as amino acids** are called as aminotelic animals.
- Aminotelic animals include some invertebrates like some molluscs (Unio, Limnaea, etc.) and some echinoderms (e.g., Asterias) excrete excess amino acids.

Classification Based On Habitat- Aquatic Or Terrestrial

Natural environments or habitat can be categorized as aquatic (water) or terrestrial (land). Depending on the habitat they live in, organisms can be categorised into aquatic or terrestrial organisms.

Aquatic animals:

Animals who **live in water for entire lifetime or most of their lifetime** are called aquatic animals. Both aquatic invertebrates and vertebrates developed entirely different adaptations to live in water unlike the animals that live on land. Aquatic animals can be divided into two broad groups depending on their aquatic habitat, namely; **marine animals** and **freshwater animals**. Some examples for aquatic invertebrates include jellyfish, corals, sea anemones, hydras, etc. These invertebrates are adapted to acquire dissolved oxygen directly from water. Aquatic vertebrates include bony fishes, cartilaginous fishes, whales, turtles, dolphins, sea lions, etc. Unlike the land animals, aquatic animals like fish, aquatic mammals have fins and streamline bodies that enable them to move fast in the water.

Terrestrial animals:

Terrestrial animals are the animals who **live on land for most or all of their life span**. Terrestrial organisms take up oxygen from the atmosphere whereas aquatic animals obtain oxygen dissolved in water. Air contains more O₂ as compared to water. Since the content of O₂ in the air is high, the terrestrial animals do not have to breathe faster to get more oxygen. Therefore, unlike aquatic animals, terrestrial animals do not need adaptations for gaseous exchange and can adapt to a relaxed mode of breathing.

Molecular taxonomy

Molecular taxonomy is a way of classifying living organisms by comparing their **DNA or RNA sequences** instead of just looking at physical features.

Every organism has genetic material (DNA/RNA) that carries information about its structure and functions. When scientists compare these sequences between different organisms, they can see how similar or different they are.

If two organisms have **very similar DNA/RNA**, it means they are **closely related** and share a recent common ancestor.

If their DNA/RNA is **very different**, they are **distantly related**.

For example, humans and chimpanzees have highly similar DNA, so they are placed close together in classification.

The objectives of molecular taxonomy are as follows:

1. It helps in reconstructing the genetic relationship between various species.
2. It helps in the study of evolution in a sequential manner.
3. It helps in finding the time of divergence occurring in between the various species.

Definition of Model Organism

A model organism is a non-human species that has been extensively studied and is easy to maintain and reproduce in a laboratory setting, as well as having specific experimental benefits.

Non-human species are kept in laboratories as model creatures to study and experiment with biological processes. Model organisms are used to learn more about other, more difficult-to-understand species.

Types of Model Organisms

The famous model organisms are given below.

Characteristics of *E.coli* as model organisms:

Escherichia coli is a rod-shaped (bacillus) Gram-negative bacterium that is frequently used as a model organism. Factors such as its ability to grow fast using cheap media and availability of molecular tools to perform genetic manipulations are favourable for using *E. coli* as a model organism in molecular genetics.

Characteristics of *S. cerevisiae* as model organisms:

Saccharomyces cerevisiae (commonly known as baker's yeast) is a single-celled eukaryote that is frequently used in scientific research. *S. cerevisiae* is an attractive model organism due to the fact that its genome has been sequenced, its genetics are easily manipulated, and it is very easy to maintain in the lab.

Characteristics of *D. melanogaster*. as model organisms:

Melanogaster are very small and thus, very easy to maintain. Due to their small size and minimal requirements, many *Drosophila* can be raised and tested within a small laboratory which does not have access time, space or funding. Genetic factors also make this fly an ideal model organism. *D. melanogaster* only has four pairs of chromosomes compared to 23 pairs in humans. This simplicity was one of the reasons why they were first used in genetic studies; *Drosophila* genes could be mapped easily to investigate genetic transmission.

Characteristics of *C. elegans* as model organisms:

Caenorhabditis elegans or *C. elegans* is a worm, has been extensively used as the model organism since the 1970s. These *C. elegans* are transparent, small worms with a rapid life cycle and large brood sizes. The majority of *C. elegans* are self-fertile hermaphrodites making it easy to conduct extensive experiments over multiple generations. Likewise to other model organisms, the *C. elegans* genome is also wholly sequenced, and about 60% of the human genes in *C. elegans* have an ortholog, making it ideal for studying the processes of basic molecular biology.



Characteristics of *A. thaliana* as model organisms:

Arabidopsis thaliana or *A. thaliana* is a small dicotyledonous species. It is originally adopted as a model organism because of its usefulness for genetic experiments. Important features included a short generation time, small size that limited the requirement for growth facilities, and prolific seed production through self-pollination.

Characteristics of *M. musculus* as model organisms:

The house mouse or *Mus musculus* is often used as a preferred model organism as mice and humans share on average 85 percent of similarly coded genes, mice can be used as a model to study how genetic mutations impact brain development. Changes in mouse DNA mimic changes in human DNA and vice-versa. In addition, mice exhibit behaviours that can be used as models for exploring human behaviour. They are also considered to be useful model organisms for understanding and studying human disease as they develop diseases that naturally affect immune, endocrine, nervous, skeletal, digestive, and cardiovascular systems.

